

1-Solutions-GenDistance

2026-09-01

Part I: Read in and Align the Sequences with Clustal-Omega

```
#Align sequences
dnaSeqs = readDNAStringSet("flu_seqs.fasta")
aligns = msa(dnaSeqs, method="ClustalOmega", order="input")

## using Gonnet
```

Part II: Use the Phangorn package to compare substitution models

```
forPhang = msaConvert(aligns, type="phangorn::phyDat")

mt = modelTest(forPhang, model=c("JC", "F81", "K80", "HKY", "TrN"))

## Model      df logLik  AIC      BIC
##      JC 93 -4898.331 9982.662 10488.49
##      JC+I 94 -4868.024 9924.048 10435.31
##      JC+G(4) 94 -4867.283 9922.566 10433.83
##      JC+G(4)+I 95 -4867.175 9924.35 10441.05
##      F81 96 -4858.893 9909.786 10431.93
##      F81+I 97 -4828.603 9851.206 10378.79
##      F81+G(4) 97 -4827.907 9849.814 10377.39
##      F81+G(4)+I 98 -4827.816 9851.632 10384.65
##      K80 94 -4754.368 9696.735 10208
##      K80+I 95 -4723.499 9636.997 10153.7
##      K80+G(4) 95 -4722.713 9635.425 10152.13
##      K80+G(4)+I 96 -4722.583 9637.166 10159.31
##      HKY 97 -4715.238 9624.476 10152.06
##      HKY+I 98 -4684.653 9565.306 10098.33
##      HKY+G(4) 98 -4683.727 9563.455 10096.47
##      HKY+G(4)+I 99 -4683.581 9565.162 10103.62
##      TrN 98 -4713.904 9623.807 10156.83
##      TrN+I 99 -4683.188 9564.375 10102.83
##      TrN+G(4) 99 -4682.404 9562.808 10101.27
##      TrN+G(4)+I 100 -4682.289 9564.579 10108.48

write.table(mt, "modelTest_results.txt", quote=FALSE, sep="\t", row.names=FALSE, col.names=TRUE)
```

Question 1: The Bayesian Information Criterion (BIC) is a method for comparing models that includes penalties for more complex models. With this method, the best one is the model with the LOWEST BIC score. In your test, which model is this? Describe this model in terms of its assumptions. (5 pts)

The lowest BIC is model HKY + G. “This model generalizes k80 by relaxing the assumption of equal base frequencies.” - help in R

Part III: Calculate Genetic Distance Under the Best Model

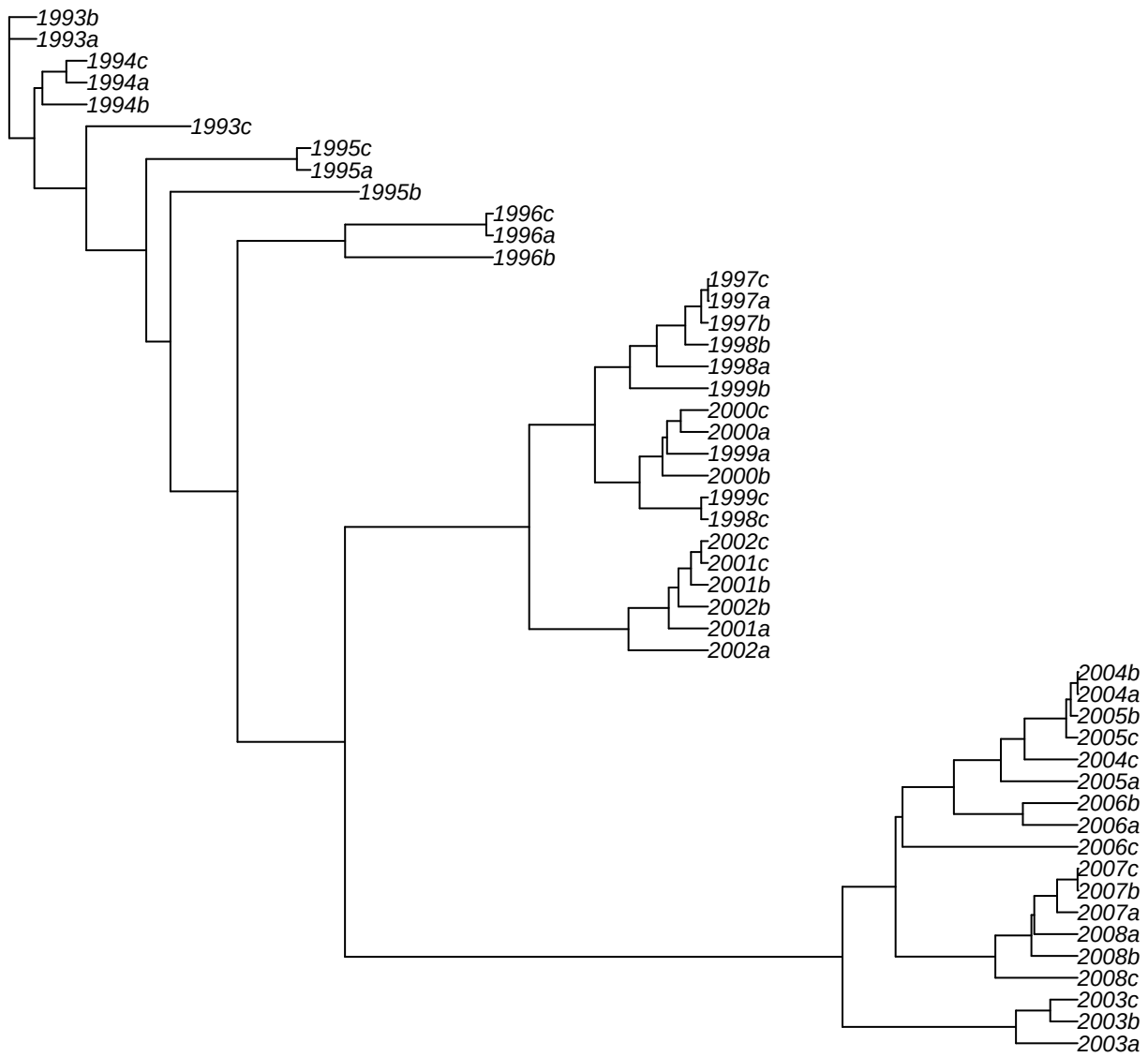
Question 2: Show your command for calculating distance, including which model you used in your command, and which option you used for gamma. (1 pt)

```
dna = as.DNABin(forPhang)
?dist.dna()
D = dist.dna(dna, model="F84", gamma=FALSE)
```

I used gamma = False. # Part IV: Create both UPGMA and Neighbor-Joining Trees

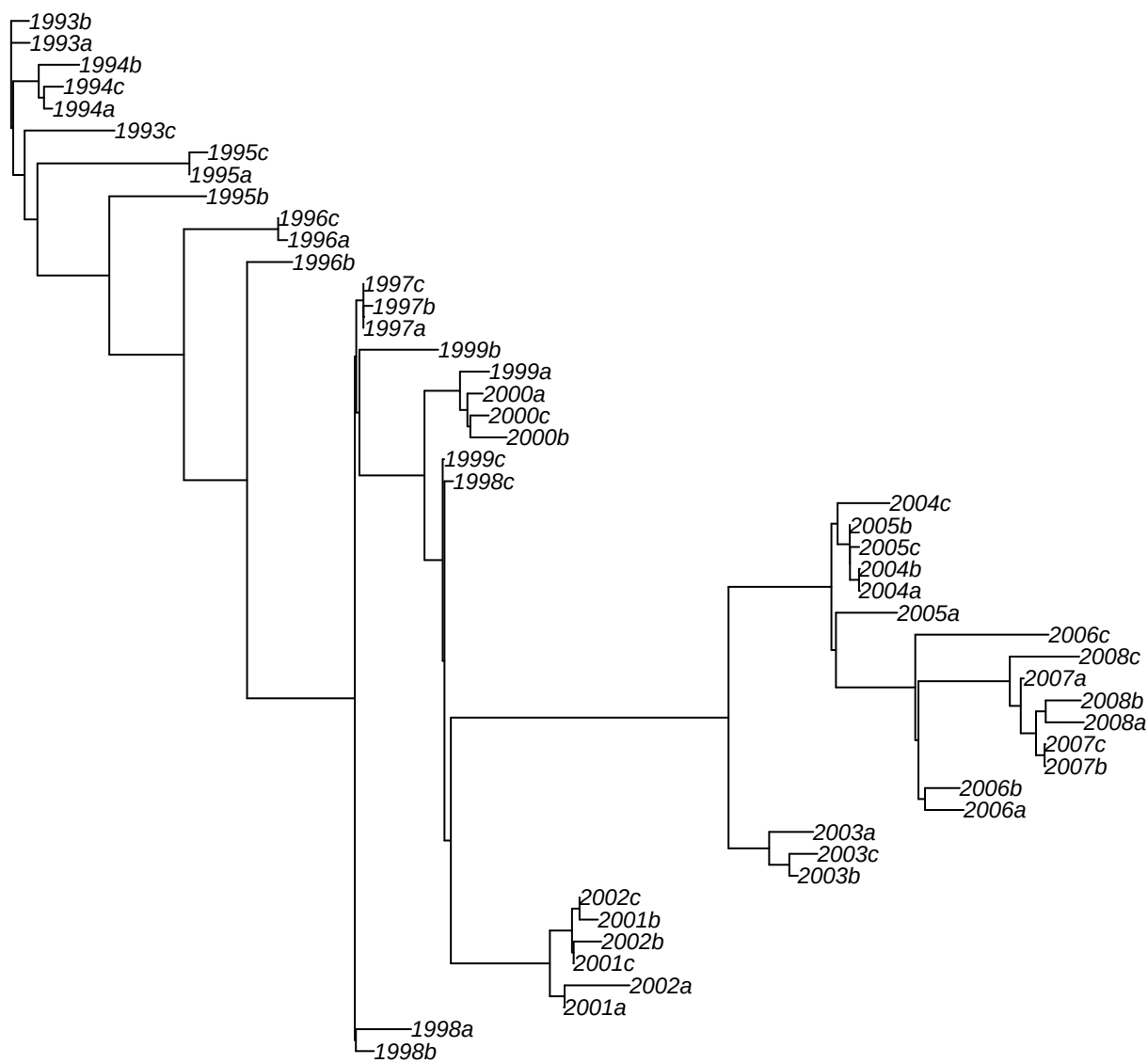
```
t.upgma = upgma(D)
t.nj = nj(D)
t.upgma.root = root(phy=t.upgma, outgroup='1993a')
t.nj.root = root(phy=t.nj, outgroup='1993a')
plot(t.upgma.root, main="UPGMA", cex=0.75)
```

UPGMA



```
plot(t.nj.root, main="Neighbor-Joining", cex=0.75)
```

Neighbor-Joining



```
treedist(t.upgma.root, t.nj.root)
```

```
##      symmetric.difference  branch.score.difference      path.difference
##                40.00000000                0.01950911      135.70556363
## quadratic.path.difference
##                0.35911714
```

Question 3: Provide your plots for the rooted UPGMA and rooted Neighbor-Joining trees (be sure they are labeled so I can tell which is which!). Describe any differences you see in the overall appearance of the two trees. Do the same clades appear in each tree? Do viruses collected in later years branch off from the viruses of previous years (like we would expect)? (5 pts)

Upgma assumes clockwise evolution which causes this symmetrical looking tree. The clades are different in both trees. Most notably after 1996. Yes, viruses in later years branch off from previous years.

Part V: Calculate Distance Metrics Between the 2 Trees

Question 4: What is the symmetric difference, branch score difference, and path difference between your two trees? Which number implies the biggest difference in the trees? Based on what you know about the definition of these metrics, why would one score be much higher than the others? Does your highest score make sense given what you see when you plot the two trees? (3 pts)

The Symmetric difference is 40. The branch score difference is 0.01950911. The Path difference is 135.70556363. The symmetric implies the biggest difference in trees. Path difference sums differences in tip-to-tip path lengths across every possible pair of taxa in the tree. Because the number of pairwise combinations grows with the square of the number of tips, it naturally produces a much larger raw number, regardless of how different the trees actually are.

Part VI: Perform Bootstrapping on the Neighbor-Joining Tree

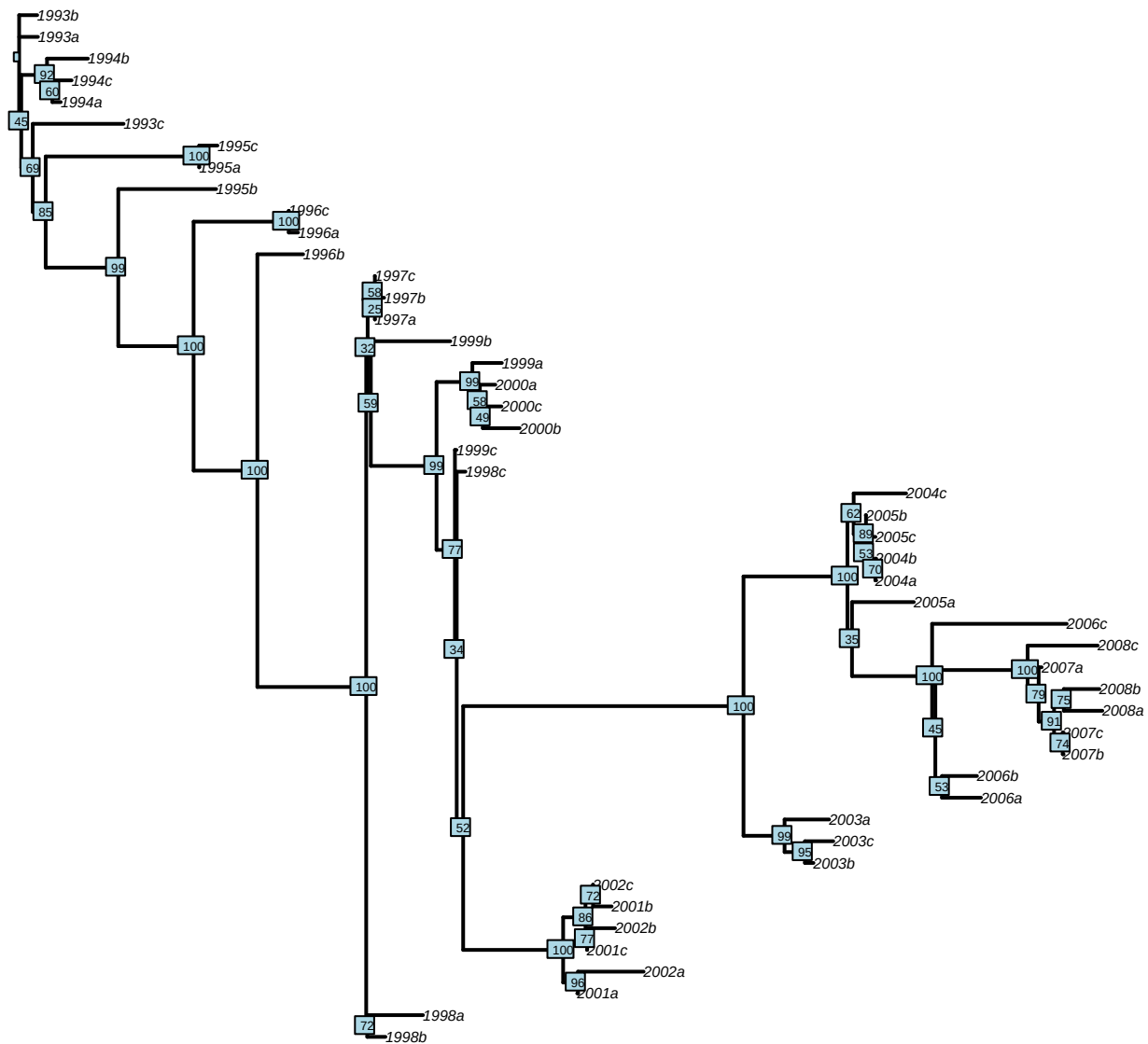
Question 5: Provide a plot of your bootstrapped tree. Are there any nodes in your tree that seem to have very weak bootstrap support (i.e. less than 50%)? How many of these weakly supported nodes do you see? Which node has the weakest support, and what clade is this node at the base of? If it is hard to see exactly where the nodes are with the bootstrap labels on them, you might also want to refer back to your original plot without the labels to help you. (3 pts)

```
myBoots = boot.phylo(t.nj.root, dna, function(x) root(nj(dist.dna(x, model="F84", gamma=FALSE)),1), tree)

## Running bootstraps:      100 / 100
## Calculating bootstrap values... done.

plot(t.nj.root, cex=0.5, edge.width=2, main="BootStrap Tree")
nodelabels(myBoots$BP, cex=0.4)
```

BootStrap Tree

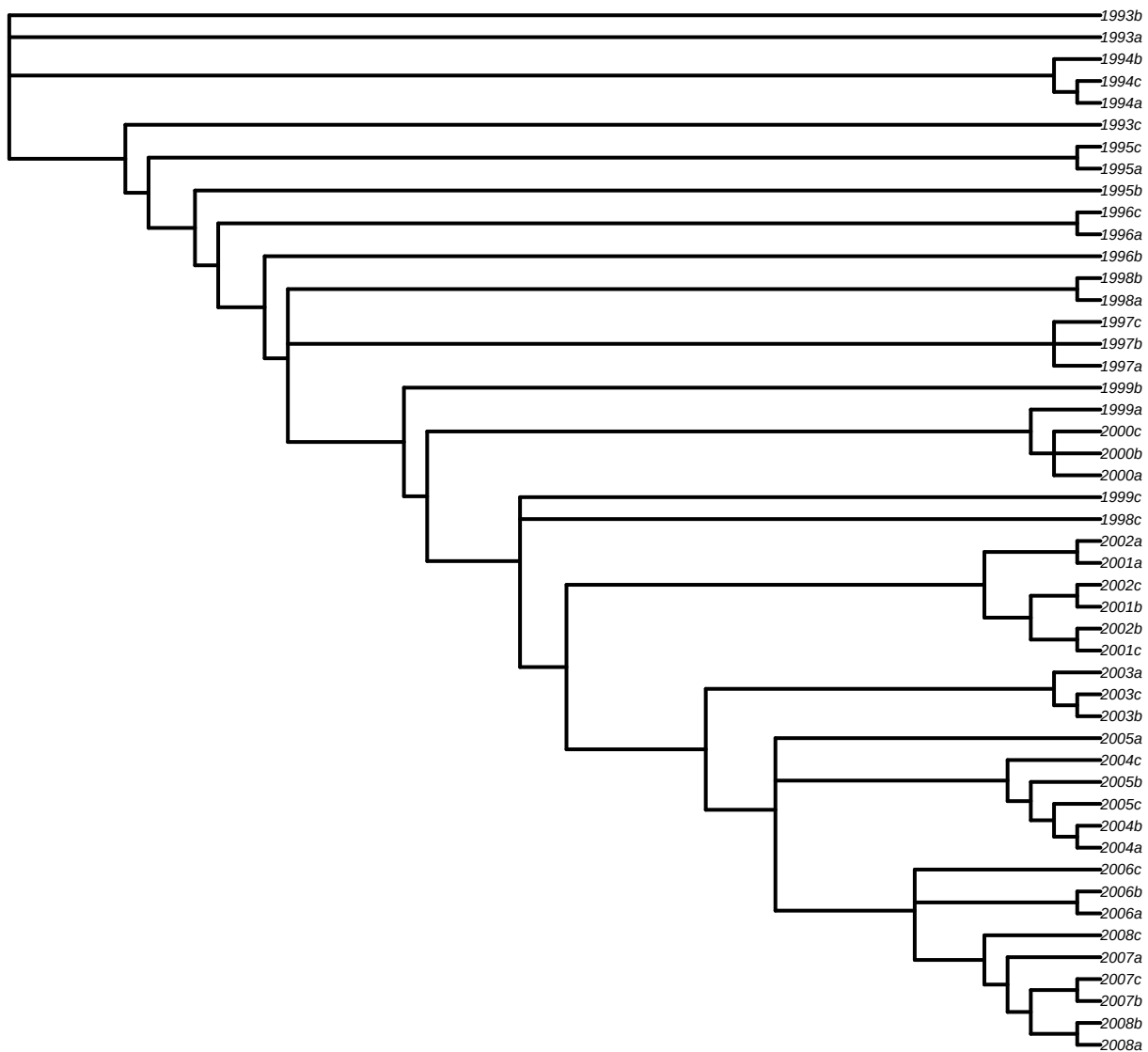


There are 7 weak nodes: 48, 47, 37, 32, 30, 26, and 23. The weakest is 23, at the node uniting all three 1997 isolates (1997a, 1997b, 1997c) into a single clade. Everytime I knit the nodes change so this number might not be accurate but the clade is.

Question 6: Provide a plot of your majority rule consensus tree. (1 pt)

```
mjtree = consensus(myBoots$trees, p=0.5)
mjtree.root = root(mjtree, outgroup='1993a')
plot(mjtree.root, cex=0.5, edge.width=2, main="Majority Rule Consensus Tree")
```

Majority Rule Consensus Tree



Part VII: Find the Most Parsimonious Tree

Question 7: What are your parsimony scores for your UPGMA and Neighbor-Joining trees? Given that a lower score is better, which of your trees is the most parsimonious? Based on what you know about the definition of parsimony, what does it mean when we say that one tree is more parsimonious than another? i.e. How should we interpret this result? (3 pts)

```
parsimony(t.upgma.root, forPhang)
```

```
## [1] 377
```

UPGMA is a score of 377 while Neighbor-Joining is a score of 340. The Neighbor-Joining is the most parsimonious. It means one tree has less independent mutations than the other when it is more parsimonious.

Question 8: Were you able to improve on the parsimony score you got with the neighbor-joining tree? (1 pt)

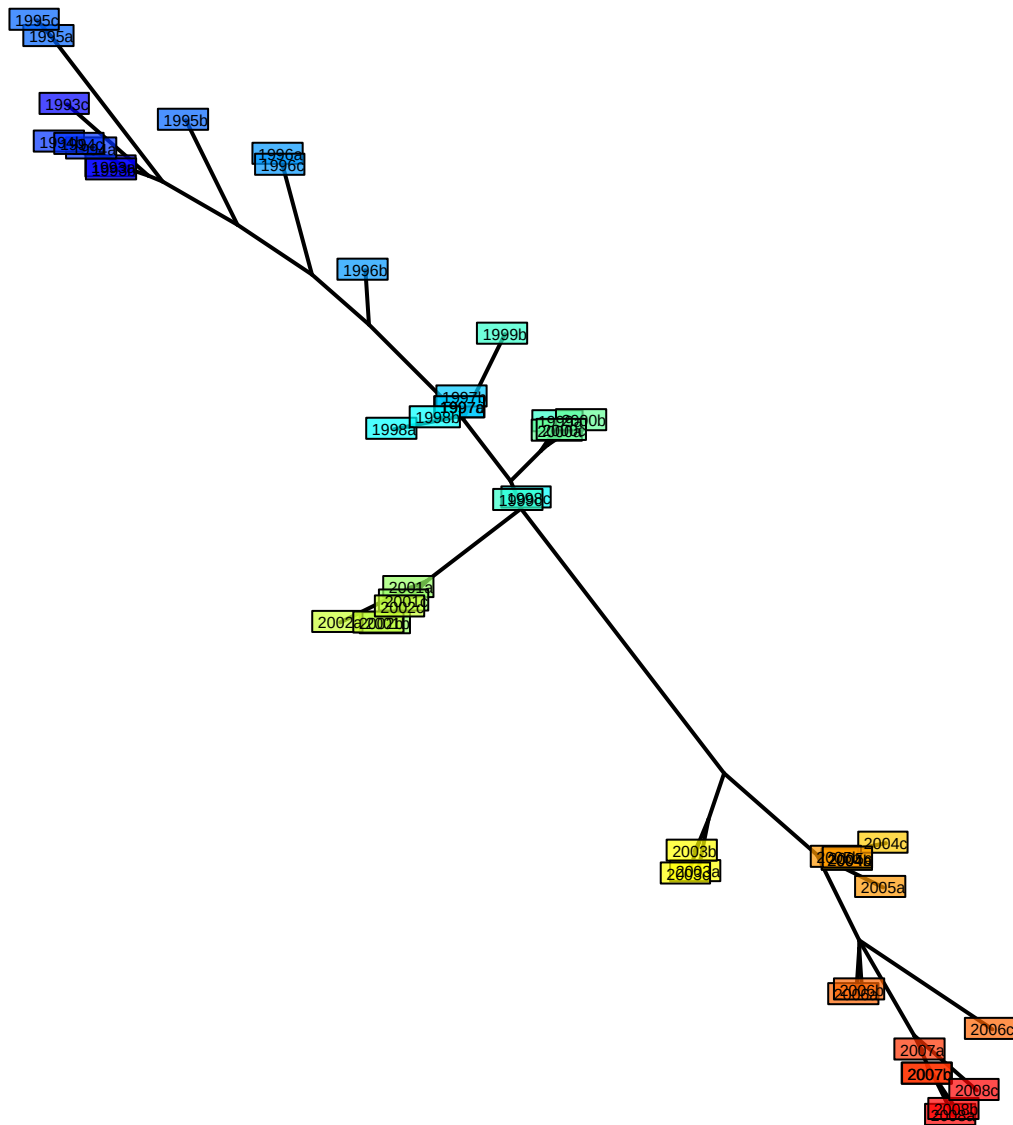
Yes

Question 9: Provide the plot for your final tree. Even though there are still some uncertain relationships, can you now see any kind of trend with respect to the years the viruses were sampled and where they appear on the tree? Describe your new tree and the trends that you see. (3 pts)

```
N <- length(t.nj.root$tip.label)
toCollapse <- match(which(myBoots$BP<50)+N, t.nj.root$edge[,2])
t.nj.root$edge.length[toCollapse] = 0
new.tree = di2multi(t.nj.root, tol=0.00001)
plot(new.tree, type="unr", show.tip=FALSE, edge.width=2)
title("Maximum-parsimony tree")
```

```
library(basetheme)
library(epicontacts)
year = as.numeric(gsub("[a-c]", "", names(forPhang)))
tiplabels(t.nj.root$tip.label, bg=transp(num2col(year),.7), cex=.5, fg="transparent")
```


Maximum-parsimony tree



The older strains are being replaced with the newer strains as time goes on. There are larger clusters in the 2004-2008 clade which contains internal branching, the rate of evolution is increasing. The shape of the tree is asymmetric.